

UME515 : Industrial Engineering

L	T	P	Cr
2	1	0	2.5

Course Objective: This course introduces the concept, tools, and techniques of industrial engineering viz. control charts, acceptance sampling, concepts of line balancing, work measurement, and production management etc., to enable the students to develop knowledge and skills in using and integrating these tools for efficient utilization of firm's resources.

Syllabus

Introduction: Industrial Engineering and its role in increasing productivity of organizations.

Plant Location & Layout: Factors affecting plant location; selection of plant site; quantitative techniques of plant location decision; principles of layout design; line balancing technique for product layout.

Quality Engineering: Variation and its types; essential dimensions of quality; seven quality tools; quality system economics; statistical quality control; applications of control charts for variables and attributes; process capability analysis; introduction to six sigma; acceptance sampling.

Production/Operations Management: Production system; types of production systems; demand forecasting and its various techniques; aggregate planning; master production scheduling; materials requirements planning; type of inventories; inventory costs; economic order quantity (under deterministic conditions); economic run length.

Work Study and Ergonomics: Method study and work measurement; principles of motion economy; principle of work sampling, MOST etc., role of ergonomics in industry, introduction to anthropometry; posture analysis; and effect of physical environment on performance.

Total Productive Maintenance (TPM): Types of maintenance; overall equipment effectiveness (OEE); pillars of TPM.

Course Learning Objectives (CLO)

The students will be able to:

1. Obtain productivity indices to evaluate effectiveness of resource utilization.
2. Analyze line balancing issues in a product layout for improving cycle times.
3. Apply quality engineering tools for process control and improvement.
4. Develop a production schedule using information/data from different functional areas.
5. Determine the time standards using work study principles/human factors in engineering.

Text Books

1. Monks, J. G., Production/Operations Management, 3rd Edition, McGraw Hill